

FUNDAMENTALS OF COMPUTER

Single-User Operating Systems

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1. The Computer System

A set of entities designed to receive, process, manage, and present information in an understandable format.

Its components are:

- **Physical components (hardware):** All the physical parts of the computer system, such as electrical or mechanical elements (keyboard, mouse, screen, hard drive, etc.) that are part of a computer.
- **Logical components (software):** The logical parts of a computer (operating system, applications, and data).
- **Human component:** People who manage, design, develop, and implement computer systems (composed of IT personnel and users).

2. Software

It is composed of all non-physical elements, mainly programs and data (operating system, applications, drivers, etc.).

Program: A set of instructions that, when executed, perform various tasks on the computer. Software is written in a programming language.

Basic software: Allows the user to use the PC without needing knowledge about the hardware or how it works internally. Examples include the operating system and drivers.

Application software: (office suites, antivirus applications, or games).

Programming software: Allows developers to write and deploy programs.

Firmware: The low-level logic that controls the electronic circuits of a device

3. Information Representation

People use numbers, letters, images, or sounds, but this information is not understandable to a machine.

An electronic circuit only recognizes the presence or absence of current, so it only understands two states: off or on (using the binary system, where 0 represents off and 1 represents on).

We must translate all information into binary so that the machine can store or interpret it.

When storing information, the computer uses powers of two, as storing it in bits would be tedious. Although powers of 10 from the decimal system have also been adopted, as the International System of Units uses that.

For this reason, the term Kilobyte was initially used to represent 1000 bytes (similar to Kg or Km), when in reality it should be 1024 bytes.

4. Operating System (OS) Concept

- A set of programs that manages the operation of the hardware and acts as an interface between it, application software, and the user.
- It allows working with the computer without having to manage the hardware directly (for example, when copying a file to the hard drive, the OS finds a free space).
- In the early computers, operating systems were not used, so it was necessary to have in-depth knowledge of the hardware. Coding was done directly in "machine language" (complex instructions in binary code), and performing simple tasks took weeks. Additionally, software only worked on computers with the same hardware. If any component changed, the software had to be rebuilt from scratch.
- Nowadays, application software will work on any computer, even if they have different hardware. We can classify operating systems based on the number of users that can work on the system where they are installed:
- **SUOS (Single-User Operating System):** Hardware resources can only be used by one user at a time. These are the OS typically used at home or by office employees (e.g., Windows 10, Windows 11, Ubuntu Desktop, or macOS).
- **MUOS (Multi-User Operating System):** Hardware resources can be used by more than one user at the same time. Each user has a monitor and keyboard connected to a central computer with a MUOS installed. Some of the most used ones are Windows systems with connected network stations.
- **NetOS (Network Operating Systems):** Hardware resources can be used by more than one user at the same time, but they don't work directly on the computer. Instead, they use a personal computer with a SUOS installed, connected to a central server with a NetOS installed. These are usually used in corporate or Internet servers (for example, a company

might have a server with a central database, a file server, etc.). The most common ones are Windows Server or Ubuntu Server.

5. History of Operating Systems

- **1940s:** First generation of computers. The computer console was used directly, and a series of micro switches were operated to load the program into the computer's memory. There were no operating systems, and users had to interact with the computer's hardware without any external assistance.
- **1950s:** To simplify interaction, operating systems made a subtle and quite simple appearance, introducing concepts such as the resident monitor (used to load programs by reading a tape or punch cards), batch processing, and temporary storage.
- **1960s:** An increase in the processing power of computers due to transistors led to techniques such as multiprogramming, time-sharing, real-time computing, and multiprocessor systems. The Atlas Supervisor and OS/360 were developed, along with MULTICS and the first editions of UNIX.
- **1970s:** The rise of minicomputers brought computing closer to the user level. Programming languages such as C and Pascal were developed. The MVS (Multiple Virtual Storage) operating system was the most widely used on mainframes.
- **1980s:** The emergence of integrated circuits (chips that contained thousands of transistors in a square centimeter of silicon) led to the rise of personal computers. Networks of PCs running NetOS and distributed systems appeared. Two of the most popular operating systems were MS-DOS, which operated on IBM 8088 CPUs and compatibles, and UNIX, on computers based on the Motorola 68000 CPU. The first versions of Apple Macintosh and Microsoft Windows also emerged.
- **1990s to Present:** Performance has increased to take advantage of hardware improvements and advancements, including increased power of microprocessors, graphic capabilities, memory size, and more.

5.1. Developed Operating Systems:

- **Windows:** The development of Windows operating systems continues. Currently, we have Windows 7, 8, 10, and 11.
- **Mac OS:** Evolution of Apple. Many versions have appeared.

- **GNU/Linux:** The first version of the Linux kernel appeared in 1991. Created by Linus Torvalds and countless collaborators over the Internet, it is based on UNIX. It operates in command mode similar to MS-DOS, although today it has a graphical user interface (GUI) with windows, thanks to a graphical server and window managers like KDE, GNOME, or CINNAMON. Recently, GNU/Linux has gained an application that converts windows into a 3D environment like Beryl, allowing users to experience Linux in a visual and appealing way. Some of the main distributions include Fedora, Debian, Ubuntu, or Linux Mint.

6. Structure of Operating Systems

Throughout history, various structures and designs have emerged. Now we will discuss the main types of architectures used in the creation of these systems:

AIX, MINIX 1 y 2, o MULTICS.

- **Monolithic Structure:** Used in early operating systems. All functionalities were implemented in a single program. The problem is that if any functionality of the system needs to be modified (such as memory management), there is a high risk of incorrectly modifying other functionalities, leading to errors. This structure was used in some of the earliest operating systems, such as UNIX and MS-DOS.
- **Layered Structure:** The operating system is divided into several layers, each responsible for a specific functionality of the system. Each layer can communicate with the upper or lower layers through service interfaces.
- This improves upon the monolithic system, as the programs that make up each layer are independent. (If something needs to be modified, such as the memory management layer, designers know that other layers will not be affected by mistake.)
- The disadvantages of designing the system in layers include how to make the division and definition of functionalities. It should be considered that the upper layer can only use the services of the immediately lower layer. Designers must be careful in planning the system for optimal performance.
- **Time expenditure:** each layer change implies additional time overhead. This type of structure has been used in operating systems such as AIX, MINIX 1 and 2, or MULTICS.
- **Modular Structure:** All modules communicate through interfaces with a central module called the Kernel. The separation into modules improves upon the monolithic structure (if one module fails, it does not affect the others). It is similar to the layered structure but is more flexible because any module can call another, not just the upper or lower ones. The modular structure is used in all current operating systems (Linux, Windows, or macOS).

7. Current Operating Systems

7.1. Windows Systems

It is the most widely used operating system today (millions of users). It has multiple versions for both personal computers and servers. There are SUOS and NetOS versions of this operating system. The current versions of Windows SUOS include Windows 10 and Windows 11. As for the NetOS, we can find Windows Server 2019 and Windows Server 2022.

Windows is distributed with commercial licenses, specifying the number of computers on which it will be installed and the intended use of the system (educational use, business use, etc).

7.2. GNU/Linux Systems

Linux is open-source, and its various implementations are free. To install it, we need to obtain a distribution. A distribution is a compilation of programs and files that are organized and ready for installation. They typically come with a GPL (GNU General Public License), which allows free access to both the binary files and the source code.

It also allows for modifications to the source code to create new distributions (under the GPL license).

Many companies, universities, and organizations have developed their own distributions since the emergence of this operating system. Some of these include:

- **Red Hat – Fedora:** Created by the company Red Hat, developing two alternative distributions for each version: a free one with GPL licensing (called Fedora Core) and a paid version (mainly for businesses) called Red Hat Enterprise Linux. Both have the same potential and tools, but the commercial version is sold with extensive documentation and technical support.
- **SuSe:** A German distribution with a commercial focus. It operates similarly to Red Hat, offering a commercial distribution (SUSE Enterprise Server) and a free version (OpenSUSE). Both have the same potential and tools, but the commercial version comes with extensive documentation and technical support.
- **Debian:** A non-commercial GNU distribution (it does not rely on any company for its development). Individual programmers collaborate to create a robust operating system. It is one of the most widespread Linux distributions, not only in business but also in home computing. There are more distributions based on Debian, such as Ubuntu, Kubuntu, and Linux Mint.

Like Windows, Linux offers SUOS and NetOS versions of its distributions. As SUOS, we can find Ubuntu Desktop or Linux Mint, and as NetOS, it is worth mentioning Debian and Ubuntu Server.

7.3. Mac OS

Mac OS (Macintosh Operating System), developed by Apple for its Macintosh computers. It was one of the first systems aimed at the public with a graphical interface featuring windows, icons, and menus.

The first version was Classic Mac OS, which was developed entirely by Apple and released in 1985. Its development extended until version 9 of the system, which was released in 1999. Since version 10 (also called Mac OS X), the system completely changed its architecture and began to be based on the UNIX operating system, although its interface retains many elements from previous versions. Mac OS X has also developed a NetOS version for servers, called Mac OS X Server. Mac OS X is also the foundation of iOS, the operating system for the iPhone and iPad, and serves as the basis for the operating system used in Apple TV.

7.4. Android

- Based on Linux, designed for mobile devices such as phones or tablets. Initially developed by Android Inc., a company initially funded by Google, which purchased it in 2005.
- It was introduced in 2007. The first mobile phone with the Android operating system was the HTC Dream (sold in October 2008).
- Android sales surpass those of Windows Phone and iOS.
- It has evolved very rapidly, with multiple versions aimed at keeping up with the technological development of the devices on which it is installed.
- Android versions are named after sweets, starting with the letter "A" and ending with "Q," due to the difficulty of finding a sweet that starts with this letter.

8. Classification of Operating Systems

8.1. By Internal Structure

This classification is based on the way operating systems are designed during their creation. According to this, we can classify the following types:

- Monolithic
- Layered
- Modular

8.2. By Number of Users

- Single User: They can only serve one user at a time (SUOS).
- Multiuser: They can serve multiple users simultaneously. Within these systems, we have MUOS and NetOS, as discussed in previous sections.

8.3. By Number of Tasks

- Single Task: Characterized by the absence of parallelism between processes (the user launches a process that prevents other functions from occurring until it is completed).
- Multitasking: Can execute multiple processes concurrently.

It can be subdivided into:

- Time-Sharing Operating Systems: Use CPU scheduling techniques to generate parallelism and alternation in the execution of processes.
- Real-Time Operating Systems: The results are correct if the calculation is correct. Processes are executed at the right moment (for example, on an automobile assembly line, tasks must be executed at the right time and done correctly).

8.4. By License System

- Proprietary: Their intellectual property belongs to a company, requiring usage licenses. Examples include Microsoft Windows and Mac OS.
- Free: Guarantee the four freedoms of software according to Richard Stallman:
 1. The freedom to use the program for any purpose.
 2. The freedom to study the program and adapt it to the user's needs.
 3. The freedom to distribute copies of the program.
 4. The freedom to improve the program and release those improvements to the public.

8.5. By Number of Processors

- Single Processor: Only allow the use of one processor.
- Multiprocessor: Allow management of multiple processors. For example:
 - Symmetric Systems: Distribute processes so that all processors have the same load.
 - Asymmetric Systems: Distribute processes to one processor or another, depending on their priority (for instance, we could have one processor for high-priority tasks and send less urgent processes to another processor).